

First-Order Modal Logic in HOL:

Deep and Shallow Embeddings with Automated Faithfulness

Christoph Benz Müller and Daniel Kirchner

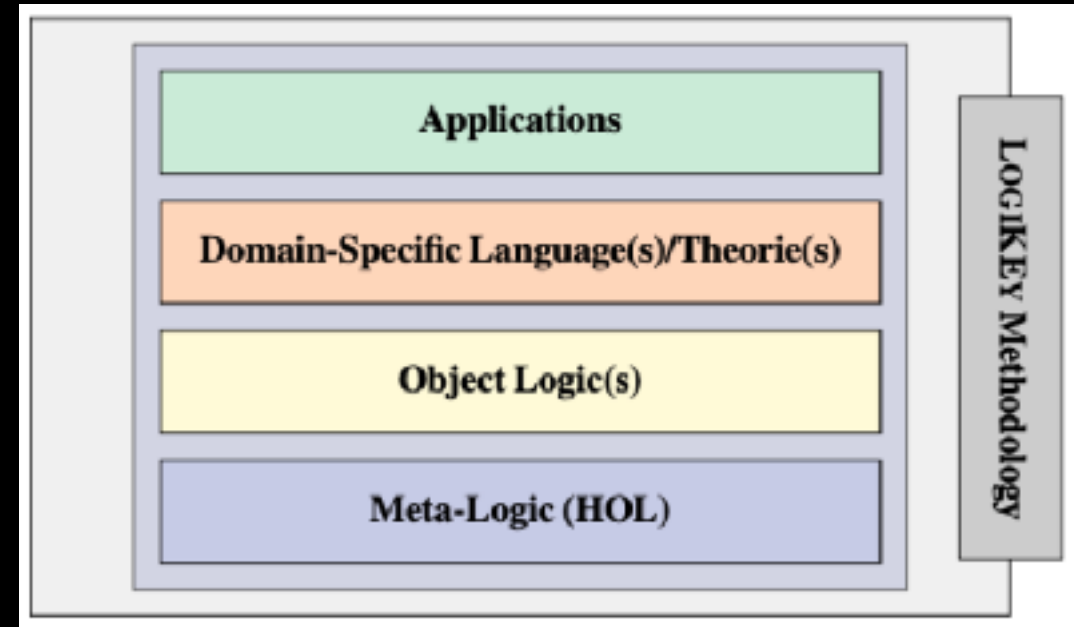
U Bamberg | FU Berlin



arXiv:2607.10880
paper + Isabelle sources

The bigger context: LogiKEy

- LogiKEy — Logic-pluralistic KR&R framework developed for more than a decade (going back to jww Larry Paulson; term LogiKEy going back to jww Xavier Parent and Leon van der Torre)
- So far primarily (minimal) shallow embeddings: the object logic as a natural fragment of HOL — good automation
- Recent work (CADE-30, 2025 — propositional): connecting shallow to deep embeddings, with automated faithfulness
- This talk: **quantified case** — FML with constant domains
 - Binders, substitution, and a Löwenheim–Skolem argument against uncountable domains



LogiKEy layered architecture



Deep vs. shallow: complementary strengths

	Deep embedding	Shallow embedding
Formulae are ...	an inductive datatype — syntax as first-class data	HOL terms (e.g. via the standard translation)
Relation to HOL	axiomatises the logic language inside HOL (datatype + recursive truth predicate)	exhibits the logic as a natural fragment of HOL — (essentially) no axioms needed
Good for ...	reasoning about the language : induction over syntax, substitution, metatheorems (e.g. Löwenheim–Skolem)	automated reasoning in/using the language , in particular, schematic problem formulations: sledgehammer proofs, nitpick countermodels
Semantics	explicit : defined by primitive recursion	implicit : inherited from HOL's own semantics

Maximal shallow = the bridge:

All semantic dependencies explicitly encoded (as in deep), yet formulae are HOL terms (as in shallow)

FML with constant-domain Kripke semantics

$\phi, \psi := r(x, y) \mid \neg\phi \mid \phi \wedge \psi \mid \exists x. \phi \mid \diamond\phi$ ($\vee, \supset, \forall, \Box$ defined as usual)

Model $\langle W, R, I, D \rangle$: worlds W , accessibility R , world-indexed interpretation I , domain D

Assignment $g: V \rightarrow D$

$\langle W, R, I, D \rangle, g, w \models r(x, y)$ iff $I(r)(w)(g(x), g(y))$

$\langle W, R, I, D \rangle, g, w \models \diamond\phi$ iff $\exists v \in W. R(w, v) \wedge \langle W, R, I, D \rangle, g, v \models \phi$

$\langle W, R, I, D \rangle, g, w \models \exists x. \phi$ iff $\exists d \in D. \langle W, R, I, D \rangle, g[x \leftarrow d], w \models \phi$

($\neg, \wedge$ as expected; validity closes over all models, g into D , $w \in W$)

- Only binary relations $\rightarrow$ n-ary definable
- Constant domain $\rightarrow$ Barcan and converse Barcan valid
- Rigid designators: g world-independent
- Basic FML of Bräuner & Ghilardi; $\exists/\diamond$ primitive

Preliminaries

Types

Variable assignments

Reflexivity, symmetry and transitivity

Notation for bounded quantifiers

$\exists w:W. \phi$

Further useful notions

```
1 theory MinFMinHOL_preliminaries imports Main
2 begin
3   <Some global settings> declare[[syntax_ambiguity_warning=false]] nitpick_params[user_axioms,expect=genuine]
4   <Declarations shared between the deep, maximal-shallow, and minimal-shallow embeddings of FML in HOL.>
5   typedecl D                               <Domain of individuals (constant across worlds).>
6   typedecl w                               <Type of possible worlds.>
7   type_synonym R = nat                     <Binary relation symbols (encoded as naturals).>
8   type_synonym V = nat                     <Individual variable symbols (encoded as naturals).>
9   type_synonym R = "D⇒D⇒bool"             <Binary relations on individuals.>
10  type_synonym W = "w⇒bool"                <Sets of possible worlds (the universe).>
11  type_synonym A = "w⇒w⇒bool"              <Accessibility relations between worlds.>
12  type_synonym I = "R⇒w⇒R"                 <World-indexed interpretations of relation symbols.>
13  type_synonym E = "V⇒D"                   <Variable assignment functions.>
14  type_synonym D = "D⇒bool"                <Domain restrictions (subsets of D as predicates).>
15  <Pointwise update of variable assignments.>
16  definition EnvMod::"E⇒V⇒D⇒E" ("_ ← _" [110,0,0] 110) where "g[x←d] = λz. if z = x then d else g z"
17  <Standard lemmas about variable-assignment update.>
18  lemma L1 [simp]: "x ≠ y ⇒ (g[y←d]) x = g x" by (simp add: EnvMod_def)
19  lemma L2: "a ≠ c ⇒ g[a←d][c←d] = g[c←d][a←d]" by (auto simp: EnvMod_def)
20  lemma L3 [simp]: "(g[a←d]) a = d" by (simp add: EnvMod_def)
21  lemma L4 [simp]: "g[a←d][a←d] = g[a←d]" by (auto simp: EnvMod_def)
22  <Standard predicates on accessibility relations (carried over from the PML preliminaries).>
23  abbreviation "reflexive" = λR::A. ∀x. R x x
24  abbreviation "symmetric" = λR::A. ∀x y. R x y ⇒ R y x
25  abbreviation "transitive" = λR::A. ∀x y z. R x y ∧ R y z ⇒ R x z
26  <Bounded quantifiers: <∀x:W. φ> stands for <∀x. W x ⇒ φ x> and <∃x:W. φ> stands for <∃x. W x ∧ φ x>.>
27  abbreviation (input) "BAll W φ" ≡ "∀x. W x ⇒ φ x"
28  syntax "BAll"::"pttrn⇒logic⇒bool⇒bool" ("({3:(_) / _})" [0,0,10]10)
29  translations "∀x:W. φ" ⇔ "CONST BAll W (λx. φ)"
30  abbreviation (input) "BEx W φ" ≡ "∃x. W x ∧ φ x"
31  syntax "BEx"::"pttrn⇒logic⇒bool⇒bool" ("({3:(_) / _})" [0,0,10]10)
32  translations "∃x:W. φ" ⇔ "CONST BEx W (λx. φ)"
33  <Set-as-predicate operations, range, and the universal domain.>
34  abbreviation (input) "Into" (infix "into" 100) where "f into D ≡ ∀x. D (f x)"
35  abbreviation (input) "Range" (infix "range" 100) where "Range f ≡ λx. ∃y. x = f y"
36  abbreviation (input) "Onto" (infix "onto" 100) where "f onto D ≡ D = Range f"
37  abbreviation (input) "Subset" (infix "⊆" 100) where "A ⊆ B ≡ ∀x. A x ⇒ B x"
38  abbreviation (input) "Union" (infixl "U" 110) where "A U B ≡ λx. A x ∨ B x"
39  abbreviation (input) "Univ" (infixl "U" 110) where "Univ = λx. True"
40  <Backward implication, useful for stating equivalences in two directions.>
41  abbreviation (input) Bimp (infixr "⇐" 50) where "φ ⇐ ψ = ψ ⇒ φ"
42 end
```

Deep embedding

How ValD and ValD' differ

$\models^d \phi$: truth in every domain D,
with assignments g ranging into $D \rightarrow$ arbitrary / varying domain.

$\models^{d'} \phi$: domain fixed to the full
universe Univ, g unrestricted
 $\rightarrow$ just the single case $D = \text{Univ}$.

Lemma Val: $\models^d \phi \Rightarrow \models^{d'} \phi$
(but: $\models^{d'} \phi \not\Rightarrow \models^d \phi$)

```

1 theory MinFMLinHOL_preliminaries imports Main
2 begin
3   <Some global settings> declare[[syntax_ambiguity_warning=false]] nitpick_params[user_axioms,expect=genuine]
4   <Declarations shared between the deep, maximal-shallow, and minimal-shallow embeddings of FML in HOL.>
5   typedecl D <Domain of individuals (constant across worlds)>
6
7 theory MinFMLinHOL_deep imports MinFMLinHOL_preliminaries
8 begin
9   <Deep embedding of (minimal) FML in HOL with constant-domain relational semantics.>
10  datatype F = AtmD R V V ("^d(_)" | NegD F ("¬^d_" [58]59) | AndD F F (infixr "^d" 56)
11    | ExD V F ("∃^d_" | DiaD F ("◇^d_" [58]59)
12
13  <Further connectives as derived definitions.>
14  definition OrD (infixr "∨^d" 54) where "φ ∨^d ψ ≡ ¬^d (¬^d φ ∧^d ¬^d ψ)"
15  definition ImpD (infixr "⊃^d" 55) where "φ ⊃^d ψ ≡ ¬^d φ ∨^d ψ"
16  definition AllD ("∀^d_" |) where "∀^d x. φ ≡ ¬^d (∃^d x. ¬^d φ)"
17  definition BoxD ("□^d_" [58]59) where "□^d φ ≡ ¬^d (◇^d (¬^d φ))"
18
19  <Relative truth in a model. Existential ranges over D; box ranges over R-successors in W.>
20  primrec RelativeTruthD::"W⇒A⇒T⇒D⇒E⇒w⇒F⇒bool" ("(_,_)_ _ ⊨^d_" [10,10,10,10,10,10] 100) where
21    "(W,R,I,D).g.w ⊨^d (r^d(x,y)) = I r w (g x) (g y)"
22    | "(W,R,I,D).g.w ⊨^d (¬^d φ) = (¬ (W,R,I,D).g.w ⊨^d φ)"
23    | "(W,R,I,D).g.w ⊨^d (φ ∧^d ψ) = ((W,R,I,D).g.w ⊨^d φ ∧ (W,R,I,D).g.w ⊨^d ψ)"
24    | "(W,R,I,D).g.w ⊨^d (∃^d x. φ) = (∃ d:D. (W,R,I,D).g[x:=d].w ⊨^d φ)"
25    | "(W,R,I,D).g.w ⊨^d (◇^d φ) = (∃ v:W. R w v ∧ (W,R,I,D).g.v ⊨^d φ)"
26
27  <Validity: true in every model, every domain-respecting assignment, and every world of the universe.>
28  definition ValD ("⊨^d_" 9) where "⊨^d φ ≡ ∀ W R I D g. g into D ⟶ (∀ w:W. (W,R,I,D).g.w ⊨^d φ)"
29
30  <Auxiliary "full-domain" notion of validity; restricts to assignments ranging over Univ of D>
31  definition ValD' ("⊨^{d'}_" 9) where "⊨^{d'} φ ≡ ∀ R I g. ∀ w. (Univ,R,I,Univ).g.w ⊨^d φ"
32
33  <General validity implies full-domain validity.>
34  lemma Val: "⊨^d φ ⟹ ⊨^{d'} φ" using ValD'_def ValD_def by simp
35
36  <Adding a nonemptiness premise on the world universe leaves validity unchanged.>
37  lemma ValD_nonemptyW: "(⊨^d φ) = (∀ W R I D g. (∃ w. W w) ⟶ g into D ⟶ (∀ w:W. (W,R,I,D).g.w ⊨^d φ))"
38  by (smt (verit, ccfv_threshold) ValD_def)
39
40  <Bag of definitions for collective unfolding.>
41  named_theorems DefD lemmas [DefD] = OrD_def ImpD_def AllD_def BoxD_def ValD_def ValD'_def
42 end
  
```

Maximal shallow embedding

```

1 theory MinFMLinHOL_shallow imports MinFMLinHOL_preliminaries
2 begin
3 —<Maximal (heavyweight) shallow embedding of FML in HOL; FML formulas are HOL terms of the following type  $\sigma$ .>
4 type_synonym  $\sigma$  = " $\mathcal{W} \Rightarrow \mathcal{A} \Rightarrow \mathcal{I} \Rightarrow \mathcal{D} \Rightarrow \mathcal{E} \Rightarrow \mathcal{W} \Rightarrow \text{bool}$ "
5 —<The five primitive cases.>
6 definition AtmS :: " $\mathcal{R} \Rightarrow \mathcal{V} \Rightarrow \mathcal{V} \Rightarrow \sigma$ " ("_s(_)" ) where " $r^s(x,y) = \lambda W R I D g w. I r w (g x) (g y)$ "
7 definition NegS :: " $\sigma \Rightarrow \sigma$ " ("¬s_" [58]59) where " $\neg^s \varphi = \lambda W R I D g w. \neg(\varphi W R I D g w)$ "
8 definition AndS :: " $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixr " $\wedge^s$ " 56) where " $\varphi \wedge^s \psi = \lambda W R I D g w. \varphi W R I D g w \wedge \psi W R I D g w$ "
9 definition ExS :: " $\mathcal{V} \Rightarrow \sigma \Rightarrow \sigma$ " (" $\exists^s$ _" 53) where " $\exists^s x. \varphi = \lambda W R I D g w. \exists d:D. \varphi W R I D (g[x \leftarrow d]) w$ "
10 definition DiaS :: " $\sigma \Rightarrow \sigma$ " (" $\Diamond^s$ _" [58]59) where " $\Diamond^s \varphi = \lambda W R I D g w. \exists v:W. R w v \wedge \varphi W R I D g v$ "
11 —<Derived connectives.>
12 definition OrS :: " $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixr " $\vee^s$ " 54) where " $\varphi \vee^s \psi = \neg^s(\neg^s \varphi \wedge^s \neg^s \psi)$ "
13 definition ImpS :: " $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixr " $\supset^s$ " 55) where " $\varphi \supset^s \psi = \neg^s \varphi \vee^s \psi$ "
14 definition AllS :: " $\mathcal{V} \Rightarrow \sigma \Rightarrow \sigma$ " (" $\forall^s$ _" 53) where " $\forall^s x. \varphi = \neg^s(\exists^s x. \neg^s \varphi)$ "
15 definition BoxS :: " $\sigma \Rightarrow \sigma$ " (" $\Box^s$ _" [58]59) where " $\Box^s \varphi = \neg^s(\Diamond^s(\neg^s \varphi))$ "
16 —<Relative truth and validity (mirroring the deep embedding).>
17 definition RelTruthS :: " $\mathcal{W} \Rightarrow \mathcal{A} \Rightarrow \mathcal{I} \Rightarrow \mathcal{D} \Rightarrow \mathcal{E} \Rightarrow \mathcal{W} \Rightarrow \sigma \Rightarrow \text{bool}$ " (" $(\_,\_,\_,\_,\_,\_) \vdash^s$ _" [100,0,0,0,0,100] 100)
18 where " $(\langle W,R,I,D \rangle, g, w) \vdash^s \varphi = \varphi W R I D g w$ "
19 definition ValS :: " $\sigma \Rightarrow \text{bool}$ " (" $\models^s$ _" 9) where " $\models^s \varphi = \forall W R I D g. g \text{ into } D \longrightarrow (\forall w:W. (\langle W,R,I,D \rangle, g, w) \vdash^s \varphi)$ "
20 —<Bag of definitions.>
21 named_theorems DefS lemmas DefS_defs [DefS] = AtmS_def NegS_def AndS_def ExS_def BoxS_def OrS_def
22 ImpS_def AllS_def DiaS_def RelTruthS_def ValS_def
23 end

```

Minimal shallow embedding as a locale

```
1 theory MinFMLinHOL_shallow_minimal_locale imports MinFMLinHOL_preliminaries
2 begin
3   <Minimal (lightweight) shallow embedding of FML in HOL, packaged as a locale.>
4   locale MinS = fixes RR ::  $\mathcal{A}$  and II ::  $\mathcal{I}$  and gg ::  $\mathcal{E}$  and WW:: $\mathcal{W}$ 
5   begin
6     <Formulas as HOL terms of type  $\sigma = w \Rightarrow \text{bool}$ : only the dependency on the world is explicit.>
7     type_synonym  $\sigma = "w \Rightarrow \text{bool}"$ 
8     <Five primitive cases. ExM, AllM are HOL binders over  $V = \text{nat}$ ; atoms consult II via gg.>
9     definition AtmM :: " $R \Rightarrow V \Rightarrow V \Rightarrow \sigma$ " ("_m'(_)_") where "rm(x.y)  $\equiv \lambda w. II\ r\ w\ (gg\ x)\ (gg\ y)$ "
10    definition NegM :: " $\sigma \Rightarrow \sigma$ " ("¬m_" [58]59) where "¬mφ  $\equiv \lambda w. \neg(\varphi\ w)$ "
11    definition AndM :: " $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixr "∧m" 56) where "φ ∧m ψ  $\equiv \lambda w. \varphi\ w \wedge \psi\ w$ "
12    definition ExM :: " $(V \Rightarrow \sigma) \Rightarrow \sigma$ " (binder "∃m" 53) where "∃md. φ d  $\equiv \lambda w. \exists d. \varphi\ d\ w$ "
13    definition DiaM :: " $\sigma \Rightarrow \sigma$ " ("◇m_" [58]59) where "◇mφ  $\equiv \lambda w. \exists v:WW. RR\ w\ v \wedge \varphi\ v$ "
14    <Derived connectives.>
15    definition OrM :: " $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixr "∨m" 54) where "φ ∨m ψ  $\equiv \neg^m(\neg^m\varphi \wedge^m \neg^m\psi)$ "
16    definition ImpM :: " $\sigma \Rightarrow \sigma \Rightarrow \sigma$ " (infixr "⊃m" 55) where "φ ⊃m ψ  $\equiv \neg^m\varphi \vee^m \psi$ "
17    definition AllM :: " $(V \Rightarrow \sigma) \Rightarrow \sigma$ " (binder "∀m" 53) where "∀md. φ d  $\equiv \lambda w. \forall d. \varphi\ d\ w$ "
18    definition BoxM :: " $\sigma \Rightarrow \sigma$ " ("□m_" [58]59) where "□mφ  $\equiv \neg^m(\Diamond^m(\neg^m\varphi))$ "
19    <Relative truth and validity, with worlds restricted to the universe WW.>
20    definition RelativeTruthM :: " $w \Rightarrow \sigma \Rightarrow \text{bool}$ " ("_m =_" ) where "w  $\models^m \varphi \equiv \varphi\ w$ "
21    definition ValM :: " $\sigma \Rightarrow \text{bool}$ " ("⊨m_" 9) where "⊨m φ  $\equiv \forall w:WW. w \models^m \varphi$ "
22    <Bag of definitions.>
23    named_theorems DefM lemmas DefM_defs [DefM] = AtmM_def NegM_def AndM_def ExM_def BoxM_def OrM_def
24    ImpM_def AllM_def DiaM_def RelativeTruthM_def ValM_def
25  end
26 end
```

Three interlinked embeddings: deep & max/min shallow

Deep $(\cdot)^d$

datatype $\mathcal{F}$ of formulae

truth by primitive recursion

→ metalogical reasoning

Maximal shallow $(\cdot)^s$

$\sigma = \mathcal{W} \Rightarrow \mathcal{A} \Rightarrow \mathcal{I} \Rightarrow \mathcal{D} \Rightarrow \mathcal{E} \Rightarrow w \Rightarrow \text{bool}$

every dependency a λ -abstraction

→ standard translation, heavyweight

Minimal shallow $(\cdot)^m$

locale $\text{MinS} (\mathcal{W}, \mathcal{R}, \mathcal{I}, \mathcal{G})$

$\sigma = w \Rightarrow \text{bool}$

→ lightweight, best automation

```
1 theory MinFMLinHOL_deep imports MinFMLinHOL_preliminaries
2 begin
3   -- Deep embedding of (minimal) FML in HOL with constant-domain relational semantics.
4   datatype F = AtmD R V V ("⊢" [58] 54) | NegD F ("¬" [58] 55) | AndD F F ("∧" [58] 56)
5     | ExD V F ("∃" [58] 57) | DiaD F ("◇" [58] 58)
6   -- Further connectives as derived definitions.
7   definition OrD (infix "∨" 54) where "φ ∨ ψ ≡ ¬(¬φ ∧ ¬ψ)"
8   definition ImpD (infix "⇒" 55) where "φ ⇒ ψ ≡ ¬φ ∨ ψ"
9   definition AllD (binder "∀" 53) where "∀x. φ ≡ ¬(∃x. ¬φ)"
10  definition BoxD (binder "□" [58] 59) where "□φ ≡ ¬(◇(¬φ))"
11  -- Relative truth in a model. Existential ranges over D; box ranges over R-successors in W.
12  primrec RelativeTruthD :: "W ⇒ A ⇒ I ⇒ D ⇒ E ⇒ w ⇒ bool" ("⊨" [100,10,10,10,10] 100) where
13    "(W,R,I,D).g.w ⊨ (r(x,y)) = (r w (g x) (g y))"
14    | "(W,R,I,D).g.w ⊨ (¬φ) = (¬(W,R,I,D).g.w ⊨ φ)"
15    | "(W,R,I,D).g.w ⊨ (φ ∧ ψ) = ((W,R,I,D).g.w ⊨ φ ∧ (W,R,I,D).g.w ⊨ ψ)"
16    | "(W,R,I,D).g.w ⊨ (φ ⇒ ψ) = (∃d:D. (W,R,I,D).g[x←d].w ⊨ φ ⇒ (W,R,I,D).g[x←d].w ⊨ ψ)"
17    | "(W,R,I,D).g.w ⊨ (∃x. φ) = (∃v:W. R w v ∧ (W,R,I,D).g.v ⊨ φ)"
18    | "(W,R,I,D).g.w ⊨ (◇φ) = (∃v:W. R w v ∧ (W,R,I,D).g.v ⊨ φ)"
19  -- Validity: true in every model, every domain-respecting assignment, and every world of the universe.
20  definition ValD ("⊨" [9] 9) where "⊨ φ ≡ ∀W R I D g. g into D ⟶ (∀w:W. (W,R,I,D).g.w ⊨ φ)"
21  -- Auxiliary "full-domain" notion of validity: restricts to assignments ranging over Univ of D.
22  definition ValD_full ("⊨" [9] 9) where "⊨ φ ≡ ∀R I D g. Vw. (Univ R I Univ).g.w ⊨ φ"
23  -- General validity implies full-domain validity.
24  lemma Val: "⊨ φ ⟹ ⊨ φ" using ValD_def ValD_full_def by simp
25  -- Adding a nonemptiness premise on the world universe leaves validity unchanged.
26  lemma ValD_nonemptyW: "(⊨ φ) = (⊨W R I D g. (λw. W w) ⟶ g into D ⟶ (∀w:W. (W,R,I,D).g.w ⊨ φ))"
27  by (smt (verit, ctt, threshold) ValD_def)
28  -- Bag of definitions for collective unfolding.
29  named_theorems DefD lemmas [DefD] = OrD_def ImpD_def AllD_def BoxD_def ValD_def ValD_full_def
30 end
```

```
1 theory MinFMLinHOL_shallow imports MinFMLinHOL_preliminaries
2 begin
3   -- Maximal (heavyweight) shallow embedding of FML in HOL; FML formulas are HOL terms of the following type σ.
4   type_synonym σ = "W ⇒ A ⇒ I ⇒ D ⇒ E ⇒ w ⇒ bool"
5   -- The five primitive cases.
6   definition AtmS :: "R ⇒ V ⇒ V ⇒ σ" ("⊢" [58] 54) where "r(x,y) = λW R I D g w. (r w (g x) (g y))"
7   definition NegS :: "σ ⇒ σ" ("¬" [58] 55) where "¬φ = λW R I D g w. ¬(φ W R I D g w)"
8   definition AndS :: "σ ⇒ σ ⇒ σ" ("∧" [58] 56) where "φ ∧ ψ = λW R I D g w. φ W R I D g w ∧ ψ W R I D g w"
9   definition ExS :: "σ ⇒ σ" ("∃" [58] 57) where "∃x. φ = λW R I D g w. ∃d:D. φ W R I D (g[x←d]) w"
10  definition DiaS :: "σ ⇒ σ" ("◇" [58] 58) where "◇φ = λW R I D g w. ∃v:W. R w v ∧ φ W R I D g v"
11  -- Derived connectives.
12  definition OrS :: "σ ⇒ σ ⇒ σ" (infix "∨" 54) where "φ ∨ ψ = ¬(¬φ ∧ ¬ψ)"
13  definition ImpS :: "σ ⇒ σ ⇒ σ" (infix "⇒" 55) where "φ ⇒ ψ = ¬φ ∨ ψ"
14  definition AllS :: "σ ⇒ σ" (binder "∀" 53) where "∀x. φ = ¬(∃x. ¬φ)"
15  definition BoxS :: "σ ⇒ σ" (binder "□" [58] 59) where "□φ = ¬(◇(¬φ))"
16  -- Relative truth and validity (mirroring the deep embedding).
17  definition RelTruthS :: "W ⇒ A ⇒ I ⇒ D ⇒ E ⇒ w ⇒ bool" ("⊨" [100,0,0,0,0,100] 100)
18    where "(W,R,I,D).g.w ⊨ φ = φ W R I D g w"
19  definition ValS :: "σ ⇒ bool" ("⊨" [9] 9) where "⊨ φ = ∀W R I D g. g into D ⟶ (∀w:W. (W,R,I,D).g.w ⊨ φ)"
20  -- Bag of definitions.
21  named_theorems DefS lemmas [DefS] = AtmS_def NegS_def AndS_def ExS_def BoxS_def OrS_def
22  ImpS_def AllS_def DiaS_def RelTruthS_def ValS_def
23 end
```

```
1 theory MinFMLinHOL_shallow_minimal_locale imports MinFMLinHOL_preliminaries
2 begin
3   -- Minimal (lightweight) shallow embedding of FML in HOL, packaged as a locale.
4   locale MinS = fixes RR :: A and II :: I and GG :: E and WW :: V
5   begin
6     -- Formulas as HOL terms of type σ = w ⇒ bool; only the dependency on the world is explicit.
7     type_synonym σ = "w ⇒ bool"
8     -- Five primitive cases. ExM, AllM are HOL binders over V = nat; atoms consult II via GG.
9     definition AtmM :: "R ⇒ V ⇒ V ⇒ σ" ("⊢" [58] 54) where "r(x,y) = λw. (r w (GG x) (GG y))"
10    definition NegM :: "σ ⇒ σ" ("¬" [58] 55) where "¬φ = λw. ¬(φ w)"
11    definition AndM :: "σ ⇒ σ ⇒ σ" (infix "∧" 56) where "φ ∧ ψ = λw. φ w ∧ ψ w"
12    definition ExM :: "σ ⇒ σ" (binder "∃" 53) where "∃x. φ = λw. ∃d. φ d w"
13    definition DiaM :: "σ ⇒ σ" (binder "◇" [58] 59) where "◇φ = λw. ∃v:WW. RR w v ∧ φ v"
14  end
15  -- Derived connectives.
16  definition OrM :: "σ ⇒ σ ⇒ σ" (infix "∨" 54) where "φ ∨ ψ = ¬(¬φ ∧ ¬ψ)"
17  definition ImpM :: "σ ⇒ σ ⇒ σ" (infix "⇒" 55) where "φ ⇒ ψ = ¬φ ∨ ψ"
18  definition AllM :: "σ ⇒ σ" (binder "∀" 53) where "∀x. φ = ¬(∃x. ¬φ)"
19  definition BoxM :: "σ ⇒ σ" (binder "□" [58] 59) where "□φ = ¬(◇(¬φ))"
20  -- Relative truth and validity, with worlds restricted to the universe WW.
21  definition RelativeTruthM :: "w ⇒ bool" ("⊨" [9] 9) where "w ⊨ φ ≡ φ w"
22  definition ValM :: "σ ⇒ bool" ("⊨" [9] 9) where "⊨ φ = ∀w:WW. w ⊨ φ"
23  -- Bag of definitions.
24  named_theorems DefM lemmas [DefM] = AtmM_def NegM_def AndM_def ExM_def BoxM_def OrM_def
25  ImpM_def AllM_def DiaM_def RelativeTruthM_def ValM_def
26 end
```

faithfulness — machine-checked, largely automated

Further experiment stated and proved in all three variants (d / s / m) — the alignment is operational

Three interlinked embeddings: deep & max/min shallow

Deep $(\cdot)^d$

datatype $\mathcal{F}$ of formulae
truth by primitive recursion
→ metalogical reasoning



Maximal shallow $(\cdot)^s$

$\sigma = \mathcal{W} \Rightarrow \mathcal{A} \Rightarrow \mathcal{J} \Rightarrow \mathcal{D} \Rightarrow \mathcal{E} \Rightarrow w \Rightarrow \text{bool}$
every dependency a λ -abstraction
→ standard translation, heavyweight



Minimal shallow $(\cdot)^m$

locale $\text{MinS}(\mathcal{W}\mathcal{W}, \mathcal{R}\mathcal{R}, \mathcal{I}\mathcal{I}, \mathcal{G}\mathcal{G})$
 $\sigma = w \Rightarrow \text{bool}$
→ **lightweight, best automation**

Simple and fully automated

difficult

```
1 theory MinFMLinHOL_faithfulness_locale imports MinFMLinHOL_deep MinFMLinHOL_shallow
2   MinFMLinHOL_shallow_minimal_locale MinFMLinHOL_deep_subst_lemma MinFMLinHOL_ElementarySubstructure
3 begin
4   <Translation from deep to maximal shallow.>
5   fun DpToShS ("[]") where
6     "[r^d(x,y)] = r^s(x,y)" | "[¬^d φ] = ¬^s [φ]" | "[φ ∧^d ψ] = [φ] ∧^s [ψ]" | "[∃^d v. φ] = (∃^s v. [φ])" | "[◇^d φ] = ◇^s [φ]"
7
8
9
10  <Faithfulness deep ↔ maximal shallow. Pointwise first; global by unfolding validity.>
11  theorem FaithfulSDlem: "(⟨(W,R,I,D),g,w⟩ ⊨^s [φ]) ↔ (⟨(W,R,I,D),g,w⟩ ⊨^d φ)"
12    by (induct φ arbitrary: g w; auto simp: DefS DefD)
13  theorem FaithfulSD: "(⊨^s [φ]) ↔ (⊨^d φ)" by (simp add: FaithfulSDlem ValD_def ValS_def)
```

Faithfulness I: deep $\leftrightarrow$ maximal shallow, the easy case

Simple and fully automated



FaithfulSDlem : $\langle W, R, I, D \rangle, g, w \models^s [[\phi]] \leftrightarrow \langle W, R, I, D \rangle, g, w \models^d \phi$ — all ϕ, g, w
FaithfulSD : $\models^s [[\phi]] \leftrightarrow \models^d \phi$ — validity transfers

- One structural induction on ϕ , generalising over g and w — auto closes every case (simp: DefS, DefD)
- Both $\exists$ -cases quantify over the full domain D — no named-variable gap, no surjectivity issue
- No substitution machinery needed — the alignment is definitional

Contrast: deep $\leftrightarrow$ minimal shallow is the hard case — it needs substitution machinery and Löwenheim–Skolem

Needed: substitution machinery

- Needed for the $\exists$ -case of the **deep** $\leftrightarrow$ **minimal-shallow** faithfulness proof — not needed in propositional setting
- The standard first-order toolbox, mechanised once for the deep embedding:
 - free / bound / fresh variables; capture-avoiding substitution $[x \leftarrow z]\phi$; substitutability; SubstitutionLemma; alphabetic renaming; safe substitution $[x \leftarrow_r z]\phi$
 - size-based induction principles SInduct, QInduct for the trickier inductions

```

1 theory MinFMinHOL_deep_subst_lemma imports MinFMinHOL_deep
2 begin
3   --<Free and bound variable occurrences. The Box case is transparent.>
4   primrec is_free (infix "free_in" 900) where
5     "x free_in (r^d(u,v)) = (x = u  $\vee$  x = v)"
6     | "x free_in ( $\neg$ ^d  $\phi$ ) = (x free_in  $\phi$ )"
7     | "x free_in ( $\wedge$ ^d  $\phi$ ) = (x free_in  $\phi$   $\vee$  x free_in  $\psi$ )"
8     | "x free_in ( $\exists$ ^d y,  $\phi$ ) = (x free_in  $\phi$   $\wedge$  x  $\neq$  y)"
9     | "x free_in ( $\exists$ ^d  $\phi$ ) = (x free_in  $\phi$ )"
10  abbreviation is_not_free (infix "not_free_in" 900) where "x not_free_in  $\phi$  =  $\neg$ (x free_in  $\phi$ )"
11  fun is_bound (infix "bound_in" 900) where
12    "x bound_in (r^d(u,v)) = False"
13    | "x bound_in ( $\neg$ ^d  $\phi$ ) = (x bound_in  $\phi$ )"
14    | "x bound_in ( $\wedge$ ^d  $\phi$ ) = (x bound_in  $\phi$   $\vee$  x bound_in  $\psi$ )"
15    | "x bound_in ( $\exists$ ^d y,  $\phi$ ) = (x = y  $\vee$  x bound_in  $\phi$ )"
16    | "x bound_in ( $\exists$ ^d  $\phi$ ) = (x bound_in  $\phi$ )"
17  abbreviation is_not_bound (infix "not_bound_in" 900) where "x not_bound_in  $\phi$  =  $\neg$ (x bound_in  $\phi$ )"
18  abbreviation not_in (infix "not_in" 900) where "x not_in  $\phi$  = x not_free_in  $\phi$   $\wedge$  x not_bound_in  $\phi$ "
19  --<A fresh variable: strictly larger than every variable occurring in  $\phi$ .>
20  primrec fresh ("fresh" '()) where
21    "fresh (r^d(u,v)) = max (u+1) (v+1)"
22    | "fresh ( $\neg$ ^d  $\phi$ ) = fresh  $\phi$ "
23    | "fresh ( $\wedge$ ^d  $\phi$ ) = max (fresh  $\phi$ ) (fresh  $\psi$ )"
24    | "fresh ( $\exists$ ^d y,  $\phi$ ) = max (x+1) (fresh  $\phi$ )"
25    | "fresh ( $\exists$ ^d  $\phi$ ) = fresh  $\phi$ "
26  lemma L5: "x bound_in  $\phi$   $\implies$  x < (fresh  $\phi$ )" by (induct  $\phi$ ) auto
27  lemma L6: "x free_in  $\phi$   $\implies$  x < (fresh  $\phi$ )" by (induct  $\phi$ ) auto
28  lemma L7: "(fresh  $\phi$ ) not_in  $\phi$ " using L5 L6 by blast
29  lemma L8: "max (fresh  $\phi$ ) (fresh  $\psi$ ) not_free_in  $\phi$ " by (metis L6 L7 max.absorb3 max_def)
30  lemma L9: "max (fresh  $\phi$ ) (fresh  $\psi$ ) not_bound_in  $\phi$ " by (metis L5 L7 max.absorb3 max_def)
31  --<Irrelevance lemma: updating a non-free variable does not affect truth.>
32  lemma L12: "y not_free_in  $\phi$   $\implies$  ((W,R,I,U),g(y $\leftarrow$ c),w  $\models$ ^d  $\phi$ ) = ((W,R,I,U),g,w  $\models$ ^d  $\phi$ )"
33  by (induct  $\phi$  arbitrary: g w) simp; metis L4 L2
34  --<Variable-for-variable substitution. Box descends transparently.>
35  primrec Subst ("[" _ "](" '())" where
36    "[x $\leftarrow$ z](r^d(u,v)) = r^d(if x = u then z else u, if x = v then z else v)"
37    | "[x $\leftarrow$ z]( $\neg$ ^d  $\phi$ ) =  $\neg$ ^d([x $\leftarrow$ z] $\phi$ )"
38    | "[x $\leftarrow$ z]( $\wedge$ ^d  $\phi$ ) = ([x $\leftarrow$ z] $\phi$ )  $\wedge$ ^d ([x $\leftarrow$ z] $\psi$ )"
39    | "[x $\leftarrow$ z]( $\exists$ ^d y,  $\phi$ ) = (if x = y then ( $\exists$ ^d y,  $\phi$ ) else ( $\exists$ ^d y, [x $\leftarrow$ z] $\phi$ ))"
40    | "[x $\leftarrow$ z]( $\exists$ ^d  $\phi$ ) =  $\exists$ ^d([x $\leftarrow$ z] $\phi$ )"
41  lemma L13 [simp]: "size [x $\leftarrow$ z] $\phi$  = size  $\phi$ " by (induct  $\phi$ ) auto
42  lemma L14 [simp]: "[x $\leftarrow$ x] $\phi$  =  $\phi$ " by (induct  $\phi$ ) auto
43  lemma L16 [simp]: assumes "a  $\neq$  x" shows "a not_free_in ([a $\leftarrow$ x] $\phi$ )" using assms by (induct  $\phi$ ) auto
44  --<Size-based induction principle (analogous to QBFs SInduct, with a Box clause).>
45  lemma SInduct:
46    assumes "A n x. P (r^d(u,v))"
47    and "A  $\phi$ . (A  $\psi$ . size  $\phi$  < size  $\psi$   $\implies$  P  $\psi$ )  $\implies$  P ( $\neg$ ^d  $\phi$ )"
48    and "A  $\phi$   $\psi$ . (A  $\chi$ . size  $\chi$  < size  $\phi$  + size  $\psi$   $\implies$  P  $\chi$ )  $\implies$  P ( $\wedge$ ^d  $\phi$ )"
49    and "A  $\phi$   $\psi$ . (A  $\chi$ . size  $\chi$   $\leq$  size  $\phi$   $\implies$  P  $\phi$ )  $\implies$  P ( $\exists$ ^d y,  $\phi$ )"
50    and "A  $\phi$ . (A  $\psi$ . size  $\psi$   $\leq$  size  $\phi$   $\implies$  P  $\psi$ )  $\implies$  P ( $\exists$ ^d  $\phi$ )"
51  shows "P  $\phi$ "
52  using assms proof (induct "size  $\phi$ " arbitrary:  $\phi$  rules: less_induct) case less thus ?case by (induct  $\phi$ ) auto qed

```

The obstacle: minimal $\exists$ reaches only Range gg

- $gg : V \rightarrow D$ is frozen as a locale parameter; the object $\exists$ is HOL's own binder — every witness is some $gg\ v$
- Plain induction gives FaithfulMDlem: $\text{deep} \Leftrightarrow \text{minimal}$ at Range gg only
- Range gg is countable ($V = \text{nat}$), D may be uncountable — full-domain faithfulness does not follow

Worked example — $\exists x. r(x, y)$

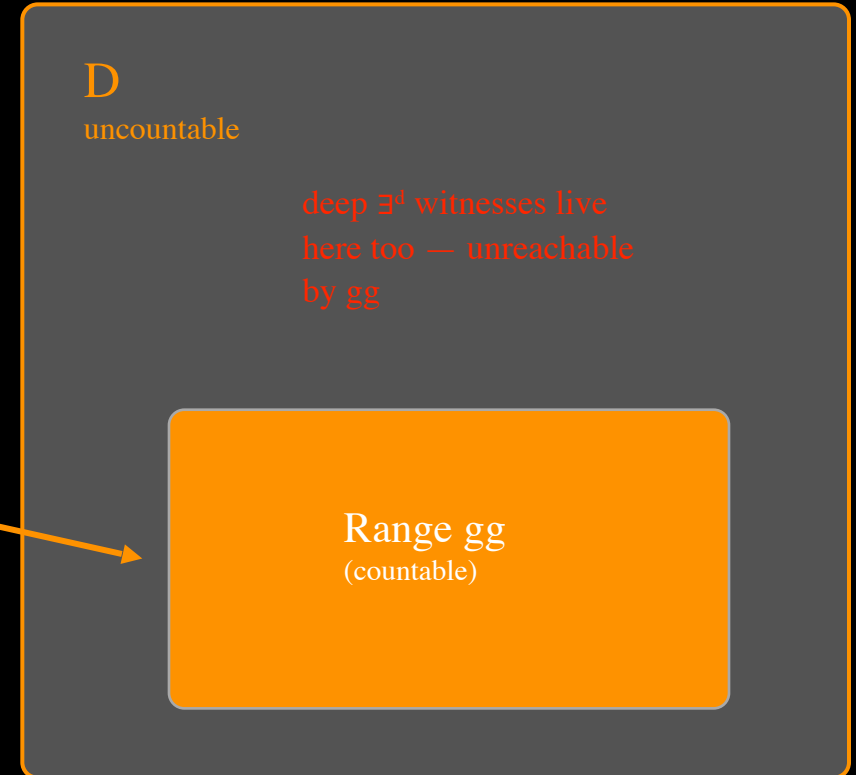
deep: true at w iff $\exists d \in D. (\Pi r\ w)\ d\ (gg\ y)$ — the witness d roams **all** of D

minimal: $\langle \exists x. r(x, y) \rangle = \lambda w. \exists v :: V. (\Pi r\ w)\ (gg\ v)\ (gg\ y)$

i.e. true at w iff $\exists a \in \text{Range } gg. (\Pi r\ w)\ a\ (gg\ y)$ — a roams only **Range** gg

$V = \text{nat}$
0 1 2 ...

gg



The gap is cardinality, not encoding — classical model theory to the rescue: shrink D instead of stretching gg

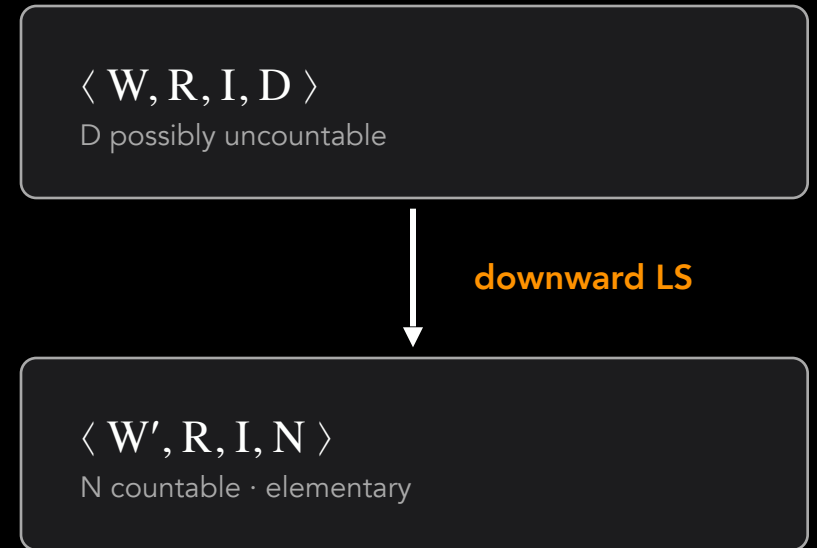
The fix: shrink the domain, don't grow the syntax

Idea

Replace D by a countable elementary N with the same truths — a nat-indexed gg can be onto N .

Downward Löwenheim–Skolem

Every model $\langle W, R, I, D \rangle$ has a countable elementary submodel $\langle W', R, I, N \rangle$ — truth preserved both ways, for all ϕ .



(elementarity thereby preserves truth for nested $\exists$ / $\diamond$ formulas — exactly the formulas that caused the problem)

Faithful at the full domain D — LS is the semantic license for minimalization while keeping $V = \text{nat}$

Downward Löwenheim–Skolem, mechanised

Every constant-domain Kripke model $\langle W, R, I, D \rangle$ with countable seeds $D_0 \sqsubseteq D$, $W_0 \sqsubseteq W$ has a countable elementary sub-model $\langle \mathcal{W}', R, I, \mathcal{N} \rangle$, where $D_0 \sqsubseteq \mathcal{N} \sqsubseteq D$ and $W_0 \sqsubseteq \mathcal{W}' \sqsubseteq W$, so that deep truth is preserved

```
255 —‹ The downward Loewenheim-Skolem theorem for FML. ›
256 theorem DownwardLowenheimSkolem:
257   assumes "D0 ⊆ D" and "W0 ⊆ W" and "countable (Collect D0)" and "countable (Collect W0)"
258   shows "∃ N W'. D0 ⊆ N ∧ N ⊆ D ∧ countable (Collect N) ∧ W0 ⊆ W' ∧ W' ⊆ W ∧ countable (Collect W')
259         ∧ (∀ g φ w. W' w → g into N → ((⟨W, R, I, D⟩, g, w ⊨d φ) = (⟨W', R, I, N⟩, g, w ⊨d φ)))"
260   apply (safe intro!: ex1[where x="N D0 W0 W R I D"] ex1[where x="W D0 W0 W R I D"])
261   apply (metis N_def D0_in_Ni)
262   apply (metis assms(1) N_subset)
263   apply (metis assms(3) assms(4) countable_N)
264   apply (simp add: W0_in_W)
265   apply (metis assms(2) W_subset)
266   apply (metis assms(4) assms(3) countable_W)
267   using N_valid assms(1,2) apply blast
268   by (metis assms(2) assms(1) N_valid)
```

- Entire file (~350 lines): Hilbert-choice witnesses for $\exists$ and $\diamond \rightarrow$ simultaneous stage-wise saturation of $\mathcal{N}$, $\mathcal{W}$
- First Löwenheim–Skolem mechanisation in the Kripke setting (to our knowledge) — reusable for other embeddings

Faithfulness, automated

- Translations $\llbracket \cdot \rrbracket$ and $\langle \cdot \rangle$
- $\text{deep} \leftrightarrow \text{maximal shallow}$
- $\text{deep} \leftrightarrow \text{minimal shallow}$: relative to Range gg

```

1 theory MinFMinHOL_faithfulness_locale imports MinFMinHOL_deep MinFMinHOL_shallow
2 MinFMinHOL_shallow_minimal_locale MinFMinHOL_deep_subst_lemma MinFMinHOL_ElementarySubstructure
3 begin
4 <Translation from deep to maximal shallow.>
5 fun DpToShS ("[]") where
6   "[ $r^d(x,y)$ ] =  $r^s(x,y)$ " | "[ $\neg^d \varphi$ ] =  $\neg^s \llbracket \varphi \rrbracket$ " | "[ $\varphi \wedge^d \psi$ ] =  $\llbracket \varphi \rrbracket \wedge^s \llbracket \psi \rrbracket$ " | "[ $\exists^d v. \varphi$ ] =  $\exists^s v. \llbracket \varphi \rrbracket$ " | "[ $\Diamond^d \varphi$ ] =  $\Diamond^s \llbracket \varphi \rrbracket$ "
7 <Translation from deep to minimal shallow (within the locale MinS.>
8 fun (in MinS) DpToShM ("[]") where
9   "[ $r^d(x,y)$ ] =  $r^m(x,y)$ " | "[ $\neg^d \varphi$ ] =  $\neg^m \langle \varphi \rangle$ " | "[ $\varphi \wedge^d \psi$ ] =  $\langle \varphi \rangle \wedge^m \langle \psi \rangle$ " | "[ $\exists^d v. \varphi$ ] =  $\langle \exists^m d. \llbracket v, d \rrbracket(\varphi) \rrbracket$ " | "[ $\Diamond^d \varphi$ ] =  $\Diamond^m \langle \varphi \rangle$ "
10 <Faithfulness deep  $\leftrightarrow$  maximal shallow. Pointwise first; global by unfolding validity.>
11 theorem FaithfulSDlem: " $\langle \langle W, R, I, D \rangle, g, w \models^s \llbracket \varphi \rrbracket \rangle \leftrightarrow \langle \langle W, R, I, D \rangle, g, w \models^d \varphi \rangle$ "
12 by (induct  $\varphi$  arbitrary; g w; auto simp: DefS DefD)
13 theorem FaithfulSD: " $\langle \models^s \llbracket \varphi \rrbracket \rangle \leftrightarrow \langle \models^d \varphi \rangle$ " by (simp add: FaithfulSDlem ValD_def ValS_def)
14 <Faithfulness deep  $\leftrightarrow$  minimal shallow, parametrised by the locale.>
15 context MinS
16 begin
17 theorem FaithfulMDlem: " $\langle w \models^m \langle \varphi \rangle \rangle \leftrightarrow \langle \langle WW, RR, II, Range \, gg \rangle, gg, w \models^d \varphi \rangle$ "
18 by (induct  $\varphi$  arbitrary; w rule: Qinduct; simp add: DefD DefM) blast
19 theorem FaithfulMD: " $\langle \models^m \langle \varphi \rangle \rangle \leftrightarrow (\forall w: WW. \langle WW, RR, II, Range \, gg \rangle, gg, w \models^d \varphi)$ "
20 using FaithfulMDlem by (simp add: ValM_def)
21 theorem FaithfulMSlem: " $\langle w \models^m \langle \varphi \rangle \rangle \leftrightarrow \langle \langle WW, RR, II, Range \, gg \rangle, gg, w \models^s \llbracket \varphi \rrbracket \rangle$ "
22 using FaithfulSDlem FaithfulMDlem by auto
23 theorem FaithfulMS: " $\langle \models^m \langle \varphi \rangle \rangle \leftrightarrow (\forall w: WW. \langle WW, RR, II, Range \, gg \rangle, gg, w \models^s \llbracket \varphi \rrbracket)$ "
24 using FaithfulMSlem by (simp add: ValM_def RelativeTruthM_def)
25 end
26 <Stronger version for elementary substructures.>

```

Global faithfulness

Quantifying over all locale interpretations of MinS recovers exactly deep validity — every locale-level proof transfers to the deep embedding in one line.

```

38 <Global faithfulness: quantifying over all locale interpretations of MinS recovers exactly deep validity.>
39 theorem FaithfulMS_all: " $\langle \forall WW \, RR \, II \, gg. \, \text{MinS.ValM } WW \, (\text{MinS.DpToShM } RR \, II \, gg \, WW \, \varphi) \rangle = \langle \models^d \varphi \rangle$ "
40 proof (unfold ValD_def; safe)
41 {
42   fix W ::  $\mathcal{W}$  and R ::  $\mathcal{A}$  and I D and g ::  $\mathcal{E}$  and x
43   assume "g into D"
44   moreover assume "W x"
45   ultimately interpret MinS_specific R I W D g  $\varphi$  x using MinS_specific_def by simp
46   assume " $\forall WW \, RR \, II \, gg. \, \text{MinS.ValM } WW \, (\text{MinS.DpToShM } RR \, II \, gg \, WW \, \varphi)$ "
47   hence " $\langle \langle WW, RR, II, Range \, gg \rangle, gg, x \models^d \varphi \rangle$ " using FaithfulMD w_in_WW by blast
48   thus " $\langle \langle W, R, I, D \rangle, g, x \models^d \varphi \rangle$ " using valid_specific by simp
49 }
50 qed (metis (mono_tags, lifting) MinS.FaithfulMD)

```

Experiments & Automation+: Prove once, transfer for free

- K, necessitation, Barcan & converse
Barcan: valid under constant domain
— uniform one-line tactics, in all three embeddings
- nitpick disproves invalid schemata with small countermodels: modal collapse $\phi \supset \Box \phi$ (2 worlds), $\Box \phi \supset \phi$ without reflexivity (T correspondence)

```

32 —<Existential variants of Barcan: under constant domain,  $\Diamond \exists$  and  $\exists \Diamond$  coincide.>
33 lemma Barcan_Dia_d: " $\models^d (\Diamond^d (\exists^d x. P^d(x,x))) \supset^d (\exists^d x. \Diamond^d P^d(x,x))$ " unfolding DefD by auto
34 lemma Barcan_Dia_s: " $\models^s (\Diamond^s (\exists^s x. P^s(x,x))) \supset^s (\exists^s x. \Diamond^s P^s(x,x))$ " unfolding DefS by auto
35 lemma Barcan_Dia_m: " $\models^m (\Diamond^m (\exists^m x. P^m(x,x))) \supset^m (\exists^m x. \Diamond^m P^m(x,x))$ " unfolding DefM by auto
36 lemma Barcan_Dia_d_conv: " $\models^d (\exists^d x. \Diamond^d P^d(x,x)) \supset^d \Diamond^d (\exists^d x. P^d(x,x))$ " unfolding DefD by auto
37 lemma Barcan_Dia_s_conv: " $\models^s (\exists^s x. \Diamond^s P^s(x,x)) \supset^s \Diamond^s (\exists^s x. P^s(x,x))$ " unfolding DefS by auto
38 lemma Barcan_Dia_m_conv: " $\models^m (\exists^m x. \Diamond^m P^m(x,x)) \supset^m \Diamond^m (\exists^m x. P^m(x,x))$ " unfolding DefM by auto

```

```

43 —<Modal collapse  $\phi \supset \Box \phi$  fails. (Note: needs full domain D for  $\models^d$ .)>
44 lemma " $\models^d P^d(x,x) \supset^d \Box^d P^d(x,x)$ " unfolding DefD apply simp nitpick oops
45 lemma " $\models^m P^m(x,x) \supset^m \Box^m P^m(x,x)$ " unfolding DefM nitpick oops
46 —<T-axiom  $\Box \phi \supset \phi$  fails without reflexivity. (Note: needs full domain D.)>
47 lemma " $\models^d \Box^d P^d(x,x) \supset^d P^d(x,x)$ " unfolding DefD apply simp nitpick oops
48 lemma " $\models^m \Box^m P^m(x,x) \supset^m P^m(x,x)$ " unfolding DefM nitpick oops

```

Transfer in one line

A theorem proved in every interpretation of the minimal-shallow locale is exported to deep validity by FaithfulMS_all — one `using MinS\_to\_Deep` discharges the deep goal, no re-proof.

```

1 theory MinFMLinHOL_experiments_locale imports MinFMLinHOL_faithfulness_locale
2 begin
3 —<Demonstration: minimal-shallow proofs that hold in every interpretation of the
4 MinS locale transfer directly to validity in the deep embedding.>
5 —<Derived rule: from validity in every minimal interpretation to deep validity.>
6 lemmas MinS_to_Deep = FaithfulMS_all[THEN iffD1, rule_format]
7 —<Additional simplification rules for derived connectives under DpToShM.>
8 lemma (in MinS) OrM_simp[DefM]: " $(\phi \vee^m \psi) = (\phi) \vee^m (\psi)$ " by (simp add: OrD_def OrM_def)
9 lemma (in MinS) ImpM_simp[DefM]: " $(\phi \supset^m \psi) = (\phi) \supset^m (\psi)$ " by (simp add: ImpD_def DefM)
10 lemma (in MinS) BoxM_simp[DefM]: " $(\Box^m \phi) = \Box^m (\phi)$ " by (simp add: BoxD_def DefM)
11 —<Example: the K axiom proved using only minimal-shallow infrastructure.>
12 lemma (in MinS) K_MinS: " $\models^m (\Box^d (P^d(x,y) \supset^d P^d(y,x)) \supset^d (\Box^d P^d(x,y) \supset^d \Box^d P^d(y,x)))$ " by (auto simp: DefM)
13 —<Transfer to the deep embedding requires no further work.>
14 lemma " $\models^d \Box^d (P^d(x,y) \supset^d P^d(y,x)) \supset^d (\Box^d P^d(x,y) \supset^d \Box^d P^d(y,x))$ " using MinS_to_Deep[OF MinS.K_MinS],
15 end

```

Conclusion — what's new here

- The deep-and-shallow methodology scales from PML to FML — binding and modality together
- Locale-packaged minimal shallow embedding + global faithfulness theorem over all interpretations
- Downward Löwenheim–Skolem for (constant-domain) Kripke semantics, surjectivity obstacle solved
- For LogiKEy: deep—maximal shallow—minimal shallow ... now also for quantified modal logics

Future work: varying domains; multimodal & non-normal modalities; other classical and non-classical logics
(well, we just did MSO in HOL using similar techniques)

MinFMLinHOL — paper, appendix, full Isabelle/HOL development (11 theory files + ROOT)
arXiv:2607.10880 · loads cleanly in current Isabelle/HOL



A6 · Related work in detail

- Benz Müller & Paulson: minimal shallow embedding of quantified/higher-order modal logic in HOL (computational metaphysics) — no parallel deep embedding; faithfulness not mechanised inside HOL
- Benz Müller & Reiche (public announcement logic): shows the need for explicit evaluation-domain dependencies in a heavyweight shallow embedding — our maximal embedding generalises that pattern
- Kirst & Shillito 2025 (Coq): completeness for first-order bi-intuitionistic logic with constant domains — a syntactic metatheorem for an intuitionistic logic; ours is a semantic deep–shallow bridge with automated proof/disproof for a classical modal one
- From 2022: Löwenheim–Skolem for FOL in Isabelle — not for constant-domain Kripke semantics, nor deployed to bridge embeddings
- Other deep embeddings of FML target metalogical results (completeness, cut-elimination) rather than mutual faithfulness with shallow embeddings
- Locales are standard for algebraic structures (Ballarín) — using them to package shallow embeddings of object logics, with a global faithfulness theorem, appears to be new

Appendix

Backup slides for questions

- A1 Shared preliminaries · locale vs. constants
- A2 Substitution machinery (sources)
- A3 Löwenheim–Skolem: the construction in detail
- A4 FaithfulMS_all: statement and proof (sources)
- A5 Experiments: Barcan, converse Barcan, nitpick countermodels
- A6 Related work in detail

A1 · Shared preliminaries · locale vs. constants

```

1 theory MinFMLinHOL_preliminaries imports Main
2 begin
3   <Some global settings> declare[[syntax_ambiguity_warning=false]] nitpick_params[user_axioms,expect=genuine]
4   <Declarations shared between the deep, maximal-shallow, and minimal-shallow embeddings of FML in HOL.>
5   typedecl D <Domain of individuals (constant across worlds).>
6   typedecl w <Type of possible worlds.>
7   type_synonym R = nat <Binary relation symbols (encoded as naturals).>
8   type_synonym V = nat <Individual variable symbols (encoded as naturals).>
9   type_synonym R' = "D => D => bool" <Binary relations on individuals.>
10  type_synonym W = "w => bool" <Sets of possible worlds (the universe).>
11  type_synonym A = "w => w => bool" <Accessibility relations between worlds.>
12  type_synonym I = "R => w => R'" <World-indexed interpretations of relation symbols.>
13  type_synonym E = "V => D" <Variable assignment functions.>
14  type_synonym D' = "D => bool" <Domain restrictions (subsets of D as predicates).>
15  <Pointwise update of variable assignments.>
16  definition EnvMod::"(D' => V => D => D)" ("[" _ " ← " _ "]" [110,0,0] 110) where "g[x ← d] ≡ λz. if z = x then d else g z"
17  <Standard lemmas about variable-assignment update.>
18  lemma L1 [simp]: "x ≠ y ⟹ (g[y ← d]) x = g x" by (simp add: EnvMod_def)
19  lemma L2: "a ≠ c ⟹ g[a ← d][c ← d] = g[c ← d][a ← d]" by (auto simp: EnvMod_def)
20  lemma L3 [simp]: "(g[a ← d]) a = d" by (simp add: EnvMod_def)
21  lemma L4 [simp]: "g[a ← d][a ← d] = g[a ← d]" by (auto simp: EnvMod_def)
22  <Standard predicates on accessibility relations (carried over from the FML preliminaries).>
23  abbreviation "reflexive" ≡ λR::A. ∀x. R x x
24  abbreviation "symmetric" ≡ λR::A. ∀x y. R x y ⟹ R y x
25  abbreviation "transitive" ≡ λR::A. ∀x y z. R x y ∧ R y z ⟹ R x z
26  <Bounded quantifiers: <∀x.W. φ> stands for <∀x. W x ⟹ φ x> and <∃x.W. φ> stands for <∃x. W x ∧ φ x>.>
27  abbreviation (input) "BAI W φ" ≡ ∀x. W x ⟹ φ x
28  syntax "BAI"::"pttrn⇒logic⇒bool⇒bool" ("["(3)(/ /) /"]" [0,0,10]10)
29  translations "∀x:W. φ" ⇔ "CONST BAI W (λx. φ)"
30  abbreviation (input) "BEF W φ" ≡ ∃x. W x ∧ φ x
31  syntax "BEF"::"pttrn⇒logic⇒bool⇒bool" ("["(3)(/ /) /"]" [0,0,10]10)
32  translations "∃x:W. φ" ⇔ "CONST BEF W (λx. φ)"
33  <Set-as-predicate operations, range, and the universal domain.>
34  abbreviation (input) "Into" (infix "into" 100) where "f into D ≡ ∀x. D (f x)"
35  abbreviation (input) "Range" where "Range f ≡ λx. ∃y. x = f y"
36  abbreviation (input) "Onto" (infix "onto" 100) where "f onto D ≡ D = Range f"
37  abbreviation (input) "Subset" (infix "⊆" 100) where "A ⊆ B ≡ ∀x. A x ⟹ B x"
38  abbreviation (input) "Union" (infixl "U" 110) where "A U B ≡ λx. A x ∨ B x"
39  abbreviation (input) "Univ" where "Univ = λx. True"
40  <Backward implication, useful for stating equivalences in two directions.>
41  abbreviation (input) Bimp (infixr "⇐" 50) where "φ ⇐ ψ ≡ ψ ⟹ φ"
42 end

```

MinFMLinHOL_preliminaries.thy

```

1 theory MinFMLinHOL_shallow_minimal imports MinFMLinHOL_faithfulness_locale
2 begin
3   <Global interpretation of the minimal shallow embedding.>
4   consts RR::A list::gg::W::V
5   global_interpretation MinS RR II gg WW.
6   <Re-introduce the locale-internal notation at the global level.>
7   notation AbmM ("[" _ " / _ "]" [58]59) and NegM ("¬" [58]59) and AndM (infixr "∧" 56) and ExM (binder "∃" 53)
8   and BoxM ("□" [58]59) and OrM (infixr "∨" 54) and ImpM (infixr "⇒" 55) and AllM (binder "∀" 53)
9   and DiaM ("◇" [58]59) and RelativeTruthM ("≡" [58]59) and ValM ("⊨" [58]59) and DpToShM ("⊨")
10 end

```

MinFMLinHOL_shallow_minimal.thy — global_interpretation

- Preliminaries shared by all embeddings: types $D, w, R; V := \text{nat}$; assignment update $g[x \leftarrow d]$ with lemmata L1–L4; reflexive/symmetric/transitive; bounded $\forall/\exists; \sqsubseteq, \sqcup$, Range, Univ, into, onto
- Locale MinS is the single source of theorems; global_interpretation with uninterpreted constants RR, II, gg, WW propagates them — no proof duplication
- Why both? nitpick handles uninterpreted constants more reliably than locale parameters; the locale enables coexisting interpretations and the global theorem

A2 · Substitution machinery (sources)

```

1 theory MinFMLinHOL_deep_subst_lemma imports MinFMLinHOL_deep
2 begin
3 --Free and bound variable occurrences. The Box case is transparent.
4 primrec is_free (infix "free_in" 900) where
5   "is_free_in (f(u,v)) = (x = u ∨ x = v)"
6   | "is_free_in (¬φ)    = (is_free_in φ)"
7   | "is_free_in (φ ∧ ψ) = (is_free_in φ ∨ is_free_in ψ)"
8   | "is_free_in (φ ∨ ψ) = (is_free_in φ ∧ is_free_in ψ)"
9   | "is_free_in (φ → ψ) = (is_free_in φ)"
10 abbreviation is_not_free (infix "not_free_in" 900) where "is_not_free_in φ ≡ ¬(is_free_in φ)"
11 fun is_bound (infix "bound_in" 900) where
12   "is_bound_in (f(u,v)) = False"
13   | "is_bound_in (¬φ)    = (is_bound_in φ)"
14   | "is_bound_in (φ ∧ ψ) = (is_bound_in φ ∨ is_bound_in ψ)"
15   | "is_bound_in (φ ∨ ψ) = (is_bound_in φ ∧ is_bound_in ψ)"
16   | "is_bound_in (φ → ψ) = (is_bound_in φ)"
17 abbreviation is_not_bound (infix "not_bound_in" 900) where "is_not_bound_in φ ≡ ¬(is_bound_in φ)"
18 abbreviation not_in (infix "not_in" 900) where "x not_in φ ≡ x not_free_in φ ∧ x not_bound_in φ"
19 --A fresh variable: strictly larger than every variable occurring in φ
20 primrec fresh ("fresh" 1) where
21   "fresh (f(u,v)) = max (u+1) (v+1)"
22   | "fresh (¬φ)    = fresh φ"
23   | "fresh (φ ∧ ψ) = max (fresh φ) (fresh ψ)"
24   | "fresh (φ ∨ ψ) = max (x+1) (fresh φ)"
25   | "fresh (φ → ψ) = fresh φ"
26 lemma L5: "x bound_in φ ⇒ x < (fresh φ)" by (induct φ) auto
27 lemma L6: "x free_in φ ⇒ x < (fresh φ)" by (induct φ) auto
28 lemma L7: "fresh φ not_in φ" using L5 L6 by blast
29 lemma L8: "max (fresh φ) (fresh ψ) not_free_in φ" by (metis L6 L7 max_absorb2 max_def)
30 lemma L9: "max (fresh φ) (fresh ψ) not_bound_in φ" by (metis L5 L7 max_absorb3 max_def)
31 --Irrelevance lemma: updating a non-free variable does not affect truth
32 lemma L12: "y not_free_in φ ⇒ ((W,R,I,D),c(y ← c),w ⊨ φ) = ((W,R,I,D),c,w ⊨ φ)"
33 by (induct φ arbitrary: c w; simp; metis L4 L12)
34 --Variable-for-variable substitution. Box descends transparently.
35 primrec Subst ("[" 1, _ "] (" 1) where
36   "[x ← z](f(u,v)) = f² (if x = u then z else u, if x = v then z else v)"
37   | "[x ← z](¬φ)    = ¬[x ← z](φ)"
38   | "[x ← z](φ ∧ ψ) = ([x ← z](φ) ∧ [x ← z](ψ))"
39   | "[x ← z](φ ∨ ψ) = (if x = y then ([x ← z](ψ)) else ([x ← z](φ)))"
40   | "[x ← z](φ → ψ) = ([x ← z](φ) → [x ← z](ψ))"
41 lemma L13 [simp]: "size [x ← z](φ) = size φ" by (induct φ) auto
42 lemma L14 [simp]: "[x ← x](φ) = φ" by (induct φ) auto
43 lemma L16 [simp]: "assumes 'x ≠ x' shows 'x not_free_in ([x ← x](φ))' using assms by (induct φ) auto
44 --Size-based induction principle (analogous to QBFs SInduct, with a Box clause).
45 lemma SInduct:
46   assumes "∀x u v. P [x ← u](u,v)"
47   and "∀φ. (∀u v. size φ < size φ ⇒ P φ) ⇒ P (¬φ)"
48   and "∀φ ψ. (∀u v. size φ < size φ + size ψ ⇒ P φ) ⇒ P (φ ∧ ψ)"
49   and "∀φ ψ. (∀u v. size φ ≤ size φ ⇒ P φ) ⇒ P (φ ∨ ψ)"
50   and "∀φ. (∀u v. size φ ≤ size φ ⇒ P φ) ⇒ P (φ → ψ)"
51   shows "P φ"
52 using assms proof (induct "size φ" arbitrary: φ rules: less_induct) case less thus ?case by (induct φ) auto qed

```

MinFMLinHOL_deep_subst_lemma.thy (excerpt)

- is_free, is_bound, fresh (with lemmata L5–L9); irrelevance lemma L12
- Variable-for-variable substitution $[x \leftarrow z]\phi$; substitutability predicate is_subst_for; SubstitutionLemma
- Alphabetic renaming ren_for_subst; safe substitution $[x \leftarrow_{\text{r}} z]\phi$ (= L27, invoked by the $\exists$ -case of the faithfulness proof)
- Size-based induction principles SInduct and QInduct
- Every modal clause is one line, dispatched by auto/force — the $\diamond$ descends transparently, like $\neg$

A3 · Löwenheim–Skolem: the construction in detail

1 Witnesses

Hilbert choice picks an individual witnessing $\exists^d y. \phi$ and an R-successor world witnessing $\diamond^d \phi$; partial assignments over the free variables of ϕ , encoded via finite lists $\rightarrow$ witness sets stay countable

2 Simultaneous saturation

Mutual recursion on stages: $\mathcal{N}_{i+1}$ adds existential witnesses, $\mathcal{W}_{i+1}$ adds modal witnesses, over assignments/worlds from stage i ; limits $\mathcal{N}$, $\mathcal{W}$ are countable unions of countable stages

3 Tarski–Vaught test

$\langle \mathcal{W}, R, I, \mathcal{N} \rangle \subseteq_e \langle W, R, I, D \rangle$: both witness conditions hold by construction; coincidence lemma lifts finite restrictions to full assignments; worlds use truth-up-to-modal-depth

Packaged for the faithfulness proofs: MinS_ES (elementary-substructure assumption), MinS_ES_Univ (universal domains), MinS_specific (per-model construction of a bounded RR and a surjective $gg \rightarrow \text{lemma valid_specific}$).

A4 · FaithfulMS_all: statement and proof (sources)

$(\forall RR \text{ II } gg \text{ WW. MinS.ValM } RR \text{ II } gg \text{ WW } (\phi)) = \models^d \phi$ — two short tactic invocations in Isabelle

```
52 MinS.ValM WW (MinS.DpToShM RR II gg WW  $\phi$ ) = ( $\models^d \phi$ )
53 proof(unfold ValD'_def; safe)
54 {
55   fix R :: A and I and g :: E and x
56   interpret MinS_specific R I Univ Univ g  $\phi$  x unfolding MinS_specific_def by simp
57   assume " $\forall WW \text{ RR II } gg. \langle WW, RR, II, Range \text{ gg} \rangle \subseteq_E \langle Univ, RR, II, Univ \rangle \longrightarrow$ "
58     MinS.ValM WW (MinS.DpToShM RR II gg WW  $\phi$ )"
59   hence "ValM (DpToShM  $\phi$ )" using ES' by blast
60   thus " $\langle Univ, R, I, Univ \rangle, g, x \models^d \phi$ " using FaithfulMD valid_specific w_in_WW by blast
61 }
62 qed(simp add: MinS_ES.FaithfulMD_ES MinS_ES_def)
63 — «Global faithfulness': quantifying over all locale interpretations of MinS_ES_Univ recovers exactly deep validity in
64   the universal domains.»
65 theorem FaithfulMS_all': " $(\forall WW \text{ RR II } gg. \text{MinS\_ES\_Univ } RR \text{ II } gg \text{ WW} \longrightarrow$ 
```

MinFMLinHOL_faithfulness_locale.thy — FaithfulMS_all_ES and FaithfulMS_all' (universal domains)

- Right-to-left: unfold deep validity, apply FaithfulMD. Left-to-right: instantiate MinS_specific at the arbitrary model, transfer via valid_specific (Löwenheim–Skolem), apply FaithfulMD
- Constants-level re-issue in MinFMLinHOL_faithfulness.thy via specification: countable seeds $DD_0, WW_0 \rightarrow$ elementary sub-model DD, WW, gg of $\langle Univ, RR, II, Univ \rangle$

A5 · Experiments: proofs and disproofs (sources)

```

27 lemma BF_s: "⊨s (∀sx. □s Ps(x,x)) ⊃s □s [∀sx. Ps(x,x)]" unfolding DefS by auto
28 lemma BF_m: "⊨m (∀mx. □m Pm(x,x)) ⊃m □m (∀mx. Pm(x,x))" unfolding DefM by auto
29 lemma CBF_d: "⊨d □d (∀dx. Pd(x,x)) ⊃d (∀dx. □d Pd(x,x))" unfolding DefD by auto
30 lemma CBF_s: "⊨s □s (∀sx. Ps(x,x)) ⊃s (∀sx. □s Ps(x,x))" unfolding DefS by auto
31 lemma CBF_m: "⊨m □m (∀mx. Pm(x,x)) ⊃m (∀mx. □m Pm(x,x))" unfolding DefM by auto
32 —<Existential variants of Barcan: under constant domain, ◇∃ and ∃◇ coincide.>
33 lemma Barcan_Dia_d: "⊨d (◇d (∃dx. Pd(x,x))) ⊃d (∃dx. ◇d Pd(x,x))" unfolding DefD by auto
34 lemma Barcan_Dia_s: "⊨s (◇s (∃sx. Ps(x,x))) ⊃s (∃sx. ◇s Ps(x,x))" unfolding DefS by auto
35 lemma Barcan_Dia_m: "⊨m (◇m (∃mx. Pm(x,x))) ⊃m (∃mx. ◇m Pm(x,x))" unfolding DefM by auto
36 lemma Barcan_Dia_d_conv: "⊨d (∃dx. ◇d Pd(x,x)) ⊃d ◇d (∃dx. Pd(x,x))" unfolding DefD by auto
37 lemma Barcan_Dia_s_conv: "⊨s (∃sx. ◇s Ps(x,x)) ⊃s ◇s (∃sx. Ps(x,x))" unfolding DefS by auto
38 lemma Barcan_Dia_m_conv: "⊨m (∃mx. ◇m Pm(x,x)) ⊃m ◇m (∃mx. Pm(x,x))" unfolding DefM by auto
39 —<Invalid schemata: nitpick reports finite countermodels.>
40 —<Symmetry of P is not implied: nitpick produces a 2-element countermodel. [Note: needs full domain D.]>
41 lemma "⊨d (∀dx. ∀dy. (Pd(x,y) ⊃d Pd(y,x)))" unfolding DefD apply simp nitpick oops

```

- Propositional tautologies, K, necessitation: auto after unfolding DefD / DefS / DefM
- Barcan & converse Barcan valid (constant domain) — uniform one-line tactics; ◇/∃ versions: ◇∃ and ∃◇ coincide
- nitpick: modal collapse $\phi \supset \Box\phi$ fails by a 2-world model; $\Box\phi \supset \phi$ fails without reflexive edges (T correspondence)
- Some disproofs use the full-domain relation $\models^{d'}$

```

42 lemma "⊨m (∀mD. ∀mV. (Pm(x,y) ⊃m Pm(y,x)))" unfolding DefM nitpick oops
43 —<Modal collapse  $\phi \supset \Box\phi$  fails. (Note: needs full domain D for  $\models^{d'}$ ).>
44 lemma "⊨d Pd(x,x) ⊃d □d Pd(x,x)" unfolding DefD apply simp nitpick oops
45 lemma "⊨m Pm(x,x) ⊃m □m Pm(x,x)" unfolding DefM nitpick oops
46 —<T-axiom  $\Box\phi \supset \phi$  fails without reflexivity. (Note: needs full domain D.)>
47 lemma "⊨d □d Pd(x,x) ⊃d Pd(x,x)" unfolding DefD apply simp nitpick oops
48 lemma "⊨m □m Pm(x,x) ⊃m Pm(x,x)" unfolding DefM nitpick oops
49 —<De-re/de-dicto distinction is collapsed under constant domain: ◇∃ implies ∃◇.>
50 lemma "⊨d ◇d (∃dx. Pd(x,x)) ⊃d (∃dx. ◇d Pd(x,x))" unfolding DefD by auto
51 lemma "⊨s ◇s (∃sx. Ps(x,x)) ⊃s (∃sx. ◇s Ps(x,x))" unfolding DefS by auto
52 lemma "⊨m ◇m (∃mx. Pm(x,x)) ⊃m (∃mx. ◇m Pm(x,x))" unfolding DefM by auto
53 end

```

MinFMLinHOL_experiments.thy (excerpts)

A6 · Related work in detail

- Benz Müller & Paulson: minimal shallow embedding of quantified/higher-order modal logic in HOL (computational metaphysics) — no parallel deep embedding; faithfulness not mechanised inside HOL
- Benz Müller & Reiche (public announcement logic): shows the need for explicit evaluation-domain dependencies in a heavyweight shallow embedding — our maximal embedding generalises that pattern
- Kirst & Shillito 2025 (Coq): completeness for first-order bi-intuitionistic logic with constant domains — a syntactic metatheorem for an intuitionistic logic; ours is a semantic deep–shallow bridge with automated proof/disproof for a classical modal one
- From 2022: Löwenheim–Skolem for FOL in Isabelle — not for constant-domain Kripke semantics, nor deployed to bridge embeddings
- Other deep embeddings of FML target metalogical results (completeness, cut-elimination) rather than mutual faithfulness with shallow embeddings
- Locales are standard for algebraic structures (Ballarín) — using them to package shallow embeddings of object logics, with a global faithfulness theorem, appears to be new

The fix: shrink the domain, don't grow the syntax

1 Idea

Replace D by a countable elementary N with the same truths — a nat-indexed gg can be onto N .

2 Downward Löwenheim–Skolem

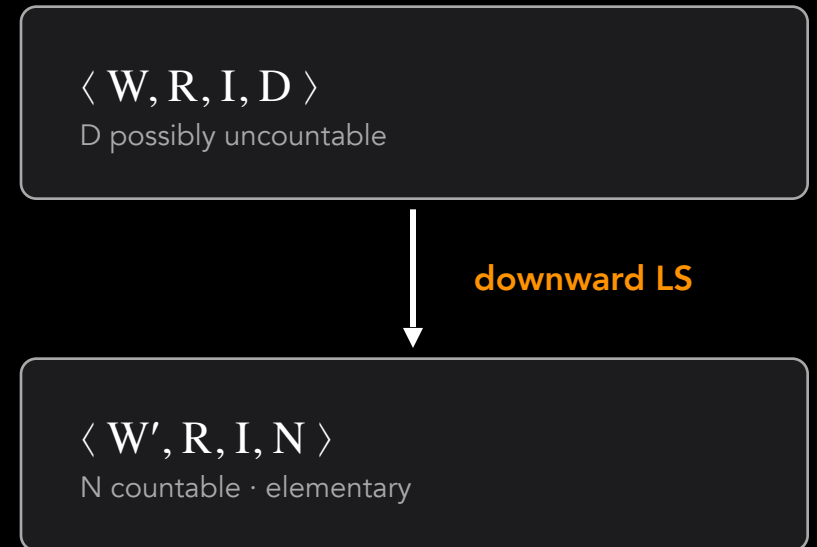
Every model $\langle W, R, I, D \rangle$ has a countable elementary sub-model $\langle W', R, I, N \rangle$ — truth preserved both ways, for all ϕ .

3 Why elementary

Transfer is prove-at- $N \rightarrow$ lift-to- D ; elementarity preserves nested $\exists$ / $\diamond$ formulas — exactly the formulas that caused the problem.

4 Tarski–Vaught

Turns “preserves all ϕ ” into “closed under $\exists$ - and $\diamond$ -witnesses” — Hilbert choice + stage-wise saturation, countable at each stage.



Faithful at the full domain D — LS is the semantic license for minimalization while keeping $V = \text{nat}$